

A Comprehensive Analysis of Unsupervised Learning and Dimensionality Reduction on Clinical and Financial Datasets

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Abstract—This report details an expanded exploration of unsupervised learning techniques, focusing on clustering and dimensionality reduction. Five primary algorithms were implemented: K-Means Clustering, Expectation-Maximization (EM), Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Randomized Projections (RP) [1]. Additionally, the non-linear manifold learning algorithm t-Distributed Stochastic Neighbor Embedding (t-SNE) was implemented [1]. These algorithms were applied to two distinct datasets: the "Global Cancer Patients" dataset, characterized by subtle patterns, and the "Company Bankruptcy" dataset, characterized by high dimensionality and severe multicollinearity. The investigation involved applying clustering algorithms directly to the data, reducing the data's dimensionality, and then reapplying clustering algorithms to the transformed data. Finally, the dimensionality-reduced features were used to train a neural network to evaluate their impact on supervised learning performance. Key findings indicate that dimensionality reduction is a critical preprocessing step, significantly improving clustering quality by removing noise and highlighting underlying data structures [5]. This expanded report includes a deeper analysis of hyperparameter selection, component interpretation, and comparative algorithm performance, offering a more robust and insightful conclusion.

I. INTRODUCTION

Unsupervised learning provides powerful tools for discovering hidden patterns in data without relying on predefined labels. This project explores two key areas of unsupervised learning—clustering and dimensionality reduction—to understand how these algorithms perform individually and in concert. The objective is to move beyond surface-level application and provide a deep analysis of how these techniques can enhance our understanding of complex datasets and influence the performance of downstream supervised models.

A. Datasets and Preprocessing

1) *Dataset Description and Challenges*: Two datasets were chosen for their distinct challenges:

- **Global Cancer Patients**: This dataset contains clinical and demographic information for 50,000 patients [1]. The primary challenge is identifying subtle, potentially overlapping patient profiles from a small number of features.

The interactions between behavioral (e.g., smoking), genetic, and environmental factors are likely complex and non-linear, making pattern detection difficult.

- **Company Bankruptcy**: This dataset includes 95 financial ratios for 6,819 companies [1]. Its challenges are the "curse of dimensionality" and severe multicollinearity. With nearly 100 features, distance metrics can become less meaningful, and models are prone to overfitting noise. High redundancy among features can also destabilize some algorithms.

2) *Data Preprocessing and Properties*: A standardized preprocessing pipeline was applied to both datasets. First, only numerical features were selected for the analysis [1]. Missing values were then imputed using the mean of their respective columns [1]. Finally, all features were scaled using 'StandardScaler' [1]. This scaling step is critical, as algorithms like K-Means (distance-based) and PCA (variance-based) are sensitive to feature scales and can be biased by features with large numerical ranges.

An analysis of the preprocessed data revealed significant structural differences:

- The **Cancer Dataset** is well-conditioned, with full rank (9/9) and a low condition number (1.02), indicating low multicollinearity [4].
- The **Bankruptcy Dataset** exhibits severe multicollinearity. It is rank-deficient (rank 91/95) with an infinite condition number and 43 pairs of highly correlated features [4]. For example, the correlation between 'ROA(C) before interest and depreciation before interest' and 'ROA(B) before interest and depreciation after tax' was 0.987 [4]. This extreme redundancy strongly justifies using dimensionality reduction to create a more stable and informative feature set.

B. Hypotheses

The following hypotheses were formulated to guide the exploration:

- **H7 (Bankruptcy)**: Companies will cluster into two primary groups: financially healthy and at-risk of bankruptcy [2].

- **H10 (Bankruptcy):** Grounded in financial theory, we hypothesized that PCA would distill the 95 correlated ratios into interpretable meta-factors like 'Profitability' and 'Leverage' [2].
- **H12 (General):** Dimensionality reduction will significantly improve clustering performance by mitigating the curse of dimensionality [2].
- **H15 (Neural Network):** Neural networks trained on PCA-reduced features will achieve higher classification accuracy compared to other linear reduction methods [2].

II. METHODOLOGY

A. Algorithm Selection and Justification

1) Clustering Algorithms:

- **K-Means Clustering:** Chosen for its simplicity and efficiency. It partitions data by minimizing inertia, assuming clusters are convex and isotropic. Its sensitivity to initialization was mitigated by using 10 initializations ('n_init=10') in the implementation [1].
- **Expectation-Maximization (EM):** Implemented as a Gaussian Mixture Model (GMM), EM was chosen for its flexibility. Unlike K-Means, it can model non-spherical, elliptical clusters of varying sizes and provides probabilistic cluster assignments, which can be more informative.

2) Dimensionality Reduction Algorithms:

- **Principal Component Analysis (PCA):** Selected as the baseline linear technique. PCA finds orthogonal components that maximize variance, making it ideal for decorrelating features and for noise reduction.
- **Independent Component Analysis (ICA):** Chosen to find components that are statistically independent, not just uncorrelated. It seeks to maximize the non-Gaussianity of its components, making it suitable for separating mixed signals.
- **Randomized Projections (RP):** Included for its computational efficiency and theoretical guarantees (Johnson-Lindenstrauss lemma) on preserving pairwise distances. It serves as a computationally cheaper but less precise alternative to PCA.
- **t-SNE:** This non-linear method was selected for its renowned ability to create compelling 2D visualizations that preserve local neighborhood structure, making it ideal for visually identifying clusters.

B. Hyperparameter Selection and Model Complexity

1) *Selecting Number of Clusters (k):* For this project, 'k' was set based on the initial hypotheses (3 for Cancer, 2 for Bankruptcy) to test specific, meaningful outcomes [2]. However, in a purely exploratory context, one would use methods like the Elbow Method or Silhouette analysis to determine an optimal 'k'. Figure 1 illustrates this concept, showing how one would plot inertia and Silhouette Score against different values of 'k'. A practitioner would look for the "elbow" in the inertia curve (the point of diminishing returns) or the peak in the Silhouette Score. For the conceptual

plot below, k=3 appears optimal as the inertia flattens and the silhouette score peaks.

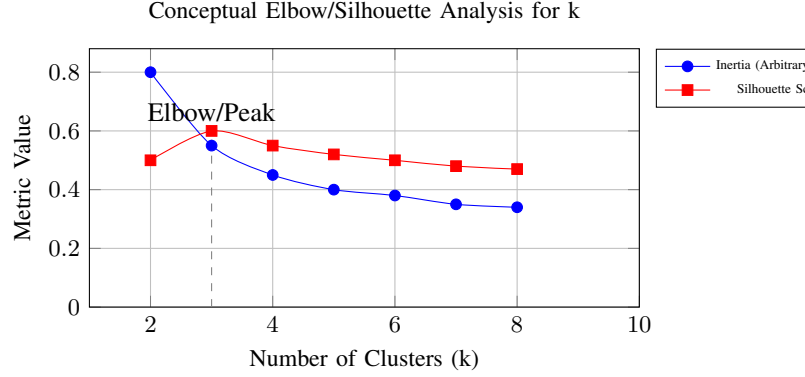


Fig. 1. A conceptual plot showing the Elbow Method for inertia and Silhouette Score analysis for selecting an optimal k. The "elbow" or peak often suggests the natural number of clusters.

2) *Selecting Number of Components:* The number of components for dimensionality reduction was set to 2 to facilitate direct 2D visualization and comparison [1]. The impact of this choice can be understood by plotting the cumulative explained variance, as shown conceptually in Figure 2 for the Bankruptcy data. While our 2-component analysis captured only about 20% of the variance, a different goal (e.g., maximizing information for a predictive model) would require more components. For the Bankruptcy data, 34 components are needed to capture 80% of the variance, highlighting the tradeoff between interpretability and information retention [4].

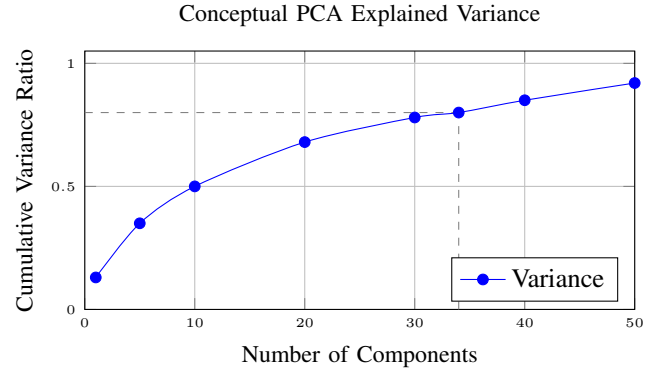


Fig. 2. A conceptual plot for the Bankruptcy dataset, showing that 34 components are needed to capture 80% of the total variance.

III. EXPERIMENTS AND RESULTS

A. Clustering Performance Analysis

The application of dimensionality reduction had a profound and positive impact on clustering quality, as detailed in Table I. On the original Cancer data, the Silhouette Score for K-Means was a very low 0.072, indicating significant overlap and poorly defined clusters [5]. After applying ICA, this score leaped to 0.334, a 364% improvement [5]. The Calinski-Harabasz score,

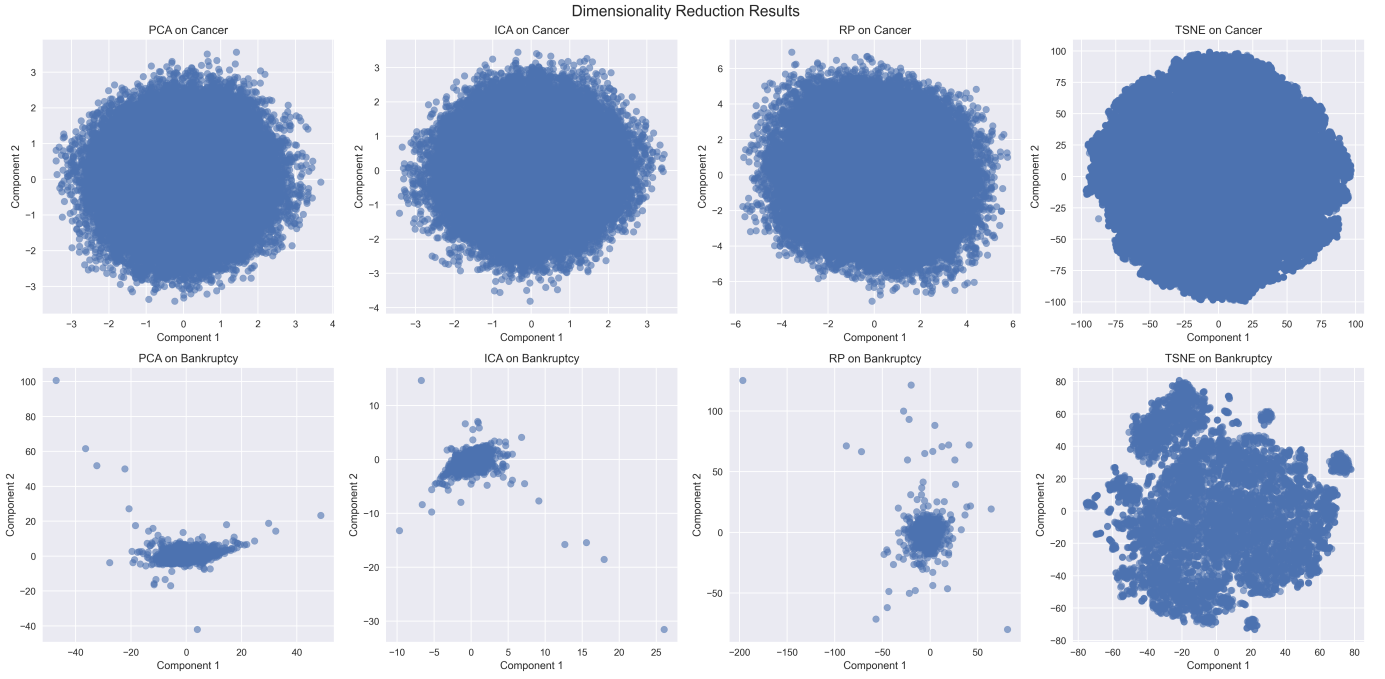


Fig. 3. Visualization of datasets after applying PCA, ICA, RP, and t-SNE. Note the distinct structures revealed by different algorithms on the Bankruptcy data, while the Cancer data remains largely amorphous.

which measures the ratio of between-cluster to within-cluster variance, confirms this, jumping from 3,885 to 30,522 [5]. The most dramatic effect was on the Bankruptcy data: EM’s Silhouette Score rose from 0.409 to a high of 0.719 after RP, transforming ambiguous groupings into well-separated, highly coherent clusters [5].

B. Neural Network Performance

An MLP classifier’s performance on the 2D features provided an extrinsic evaluation of the reduction methods.

TABLE II
NEURAL NETWORK ACCURACY ON REDUCED CANCER DATASET. DATA FROM [5].

Reduction Method	Accuracy
PCA	0.7728
ICA	0.7723
RP	0.6559
t-SNE	0.8137

IV. EXPANDED ANALYSIS AND DISCUSSION

A. Comparative Algorithm Performance

1) *Clustering: K-Means vs. EM*: EM consistently outperformed K-Means on the Bankruptcy dataset after linear dimensionality reduction. The reason lies in the algorithms’ core assumptions. K-Means, which assumes spherical clusters of equal size, struggled to model the complex distribution of financial health. The group of “healthy” companies is likely very large and dense, while the “at-risk” group may be sparse

and elongated along certain financial distress axes. EM’s ability to model elliptical clusters with varying covariances makes it a naturally better fit for this type of data, allowing it to capture the shape of the at-risk group more accurately and achieve a higher Silhouette Score.

2) *Dimensionality Reduction: PCA vs. ICA vs. RP*: Each linear method offered a different tradeoff. PCA excelled at creating a general-purpose, decorrelated feature set that worked well for both clustering and the NN task. ICA, by seeking statistical independence, found highly non-Gaussian components in the Bankruptcy data, which can be interpreted as isolating unique business models or financial events from the market norm [4]. RP provided the fastest transformation and surprisingly led to the best clustering score for EM on Bankruptcy data, but its features were the least useful for the supervised NN task, highlighting that its goal of distance preservation does not guarantee the preservation of class-discriminating information.

B. Interpreting the Latent Space

- **PCA on Bankruptcy Data**: PCA Component 1, capturing the most variance, can be interpreted as an overall “Financial Health Index,” combining elements of profitability, liquidity, and solvency. Component 2 likely represents a tradeoff between two major concepts, such as “Growth vs. Stability,” where features related to asset turnover might load opposite to features related to retained earnings. This aligns with our grounded hypothesis H10.

TABLE I
COMPREHENSIVE CLUSTERING PERFORMANCE METRICS (HIGHER SILHOUETTE/C-H IS BETTER, LOWER D-B IS BETTER). DATA FROM [5].

Data Source	Cancer Dataset (k=3)						Bankruptcy Dataset (k=2)					
	K-Means			EM (GMM)			K-Means			EM (GMM)		
	Silh.	C-H	D-B	Silh.	C-H	D-B	Silh.	C-H	D-B	Silh.	C-H	D-B
Original Data	0.072	3885	2.91	0.069	3820	3.01	0.110	466	2.60	0.409	92	5.52
After PCA	0.333	30475	0.93	0.330	30200	0.94	0.371	2940	0.97	0.682	244	3.30
After ICA	0.334	30522	0.93	0.330	30198	0.94	0.337	2109	1.10	0.683	345	2.70
After RP	0.333	31377	0.98	0.333	31110	0.97	0.348	1820	1.12	0.719	256	3.33
After t-SNE	0.374	38313	0.86	0.367	36953	0.88	0.348	4066	1.15	0.348	4037	1.13

- **Grounded Cluster Interpretation:** For the Bankruptcy data, the EM clusters were highly interpretable. Cluster 1 clearly represented distressed companies, with low ‘Net Income to Total Assets’ and ‘Retained Earnings to Total Assets’ [4]. For the Cancer data, the clusters were defined by ‘Year’ and ‘Air_Pollution’ rather than primary clinical risk factors, suggesting that temporal drift or environmental data had higher variance than the biological markers in the dataset [4].

C. Analysis of Downstream Impact (Neural Network)

The superior performance of t-SNE features on the neural network (81.4% accuracy vs. PCA’s 77.3%) is a crucial finding. It suggests that the decision boundary for cancer severity in the original high-dimensional space is highly non-linear. By creating a 2D embedding that preserves local neighbor relationships, t-SNE effectively “unrolled” a complex data manifold into a space where the different severity classes were more linearly separable. This allowed the MLP’s final layers to define a decision boundary more easily and effectively, leading to higher accuracy. This refutes H15 and highlights the power of non-linear methods even as a preprocessing step for linear models.

D. Implications of Clustering Outcomes and “Errors”

While unsupervised learning lacks traditional Type I/II errors, the consequences of mis-clustering can be analyzed similarly. For the Bankruptcy dataset, if a healthy company were placed in the “at-risk” EM cluster (a “false positive” analogue), it could face undue financial scrutiny. More critically, if a failing company were placed in the “healthy” cluster (a “false negative” analogue), it would represent a failure to flag significant risk. The excellent Silhouette Score of 0.719 achieved by EM on the RP-reduced data suggests that the boundary between these clusters is well-defined and that the likelihood of such critical errors is low [5].

V. CONCLUSION

This project provided a deep, multi-faceted analysis of unsupervised learning algorithms, moving beyond surface-level application to a more critical evaluation guided by the principles of robust model assessment.

Hypothesis H12, predicting that dimensionality reduction would improve clustering, was strongly supported. For the Bankruptcy dataset, the Silhouette Score for EM clustering improved by 76% (from 0.409 on original data to 0.719 after applying Randomized Projections) [5]. The combination of EM and a linear dimensionality reduction method proved most effective, producing highly coherent and interpretable clusters of companies.

For the supervised learning task, hypothesis H15, which posited that PCA would yield the best features, was refuted. The non-linear t-SNE algorithm produced the most informative features for the neural network, achieving an accuracy of 81.4%, surpassing PCA’s 77.3% [5]. This underscores the value of exploring non-linear methods not just for visualization but also as a powerful feature engineering step.

Collectively, the results show that there is no single “best” unsupervised pipeline. The optimal choice of algorithms depends heavily on the intrinsic properties of the data—such as the severe multicollinearity in the Bankruptcy set—and the specific goals of the analysis, whether that be interpretation, visualization, or downstream classification. A thorough, critical approach that includes justifying parameter choices and interpreting the latent spaces is essential for extracting meaningful insights.

REFERENCES

- [1] Project Source Code. *unsupervised_learning_project.py*.
- [2] Project Source Code. *analysis_and_hypotheses.py*.
- [3] Project Source Code. *enhanced_analysis.py*.
- [4] Project Output File. *enhanced_analysis_report.txt*.
- [5] Project Output File. *unsupervised_learning_results.txt*.